Price elasticity of demand (PED)

1.3.2.4 The market · Unit 1 · Mr. Akram — Akademy

PED measures how responsive demand is to a change in price. It is normally negative (ignore the minus sign when interpreting).

PED = % change in quantity demanded ÷ % change in price

- **Inelastic** (|PED| < 1) — demand changes less than proportionally.
- **Elastic** (|PED| > 1) — demand changes more than proportionally.
- **Unitary** (|PED| = 1) — demand changes exactly in proportion.

Worked example: 200 cans at £0.50; price to £0.60 cuts demand to 180. %ΔQd = –10%, %ΔP = +20%, PED = –0.5 (inelastic).

Factors affecting PED

- Availability of substitutes (more → more elastic).
- Necessity vs luxury (necessities more inelastic).
- Proportion of income (cheap items more inelastic).
- Time (more elastic in the long run).
- Brand loyalty / habit (more inelastic).

PED, pricing & total revenue

Demand is…	Raise price	Cut price
Inelastic (|PED|<1)	Revenue rises	Revenue falls
Elastic (|PED|>1)	Revenue falls	Revenue rises

%ΔQd = PED × %ΔP — e.g. PED –2 with a 25% price cut gives +50% quantity demanded. The site has an interactive PED calculator.